

Sarra MOKRANI

Computer Science Graduate - Big Data Analytics

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Profile

Computer Science graduate who has completed Master's studies in Big Data Analytics at the University of Science and Technology Houari Boumediene (USTHB). Seeking opportunities focused on data engineering, predictive analytics, SQL optimization, and business intelligence. Skilled in Python, SQL, web development, dashboards, API integration, and software design, with hands-on experience building web applications and AI-assisted data solutions.

Education

Master's Degree in Big Data Analytics

2024 – 2026

University of Science and Technology Houari Boumediene (USTHB)

Bab Ezzouar, Algiers, Algeria

Completed graduate studies focused on Big Data Analytics, data processing, predictive analytics, and decision-support systems.

Bachelor's Degree in Information Systems and Software Engineering

Sep. 2021 – Jun. 2024

University of Science and Technology Houari Boumediene (USTHB)

Bab Ezzouar, Algiers, Algeria

Coursework and practice in information systems, software engineering, databases, web development, UML modeling, and software architecture.

Baccalaureate in Mathematics

September 2018 – June 2021

Lycée Said Hamdine

Algiers, Algeria

Distinction: Highest Honors (Très Bien).

Projects

Djezzy Master's Internship Project - Intelligent Chatbot for Teradata SQL

Feb. 2026 – Jun. 2026

Master's Degree Big Data Internship

Djezzy

- Developing an AI-powered chatbot to generate, optimize, and document Teradata SQL queries from natural language questions.
- Designing a RAG-based knowledge system using database metadata, existing SQL queries, and schema context to improve query accuracy.
- Implementing query optimization support for joins, date filters, indexing recommendations, and performance-oriented SQL improvements.
- Building an interface with Streamlit/FastAPI to help engineers and business users explore telecom data without advanced SQL expertise.

Real-Time Taxi Streaming Pipeline with Kafka, MongoDB and Cassandra

2025 – 2026

Academic Big Data Project

USTHB

- Designed a real-time Big Data pipeline for NYC Yellow Taxi Trips using Apache Kafka producers, consumers, and JSON event streaming.
- Modeled and stored streaming taxi events in MongoDB sharded clusters and Cassandra multi-node clusters for comparative NoSQL analysis.
- Containerized the full architecture with Docker Compose, including Kafka multi-broker setup, MongoDB sharding, Cassandra clustering, and pipeline orchestration.
- Implemented analytical queries and benchmarking to compare ingestion throughput, query latency, horizontal scalability, and fault tolerance.

Security Web Application - Encryption, Password Attacks and Steganography

2024 – 2025

Academic Security Project

USTHB

- Developed a Python/Flask web interface covering encryption/decryption, password security, attack simulation, and text steganography modules.
- Implemented classical encryption algorithms including mirror, affine, shift, and Caesar ciphers for message exchange between two entities.
- Added decryption workflows and analyzed security weaknesses of each algorithm to highlight limitations of classical cryptography.
- Structured the application into clear modules to support demonstrations, testing, and performance evaluation of the implemented features.

Technology Watch Automation Tool - Big Data Ethics and Regulations

2024 – 2025

Academic Software Project

USTHB

- Developed a Python-based prototype to automate technology watch workflows, covering source identification, data collection, analysis, and information diffusion.
- Implemented automated information collection concepts using web scraping and API integration to retrieve articles, reports, patents, and regulatory updates.
- Designed filtering and analysis features based on keywords, statistics, trends, and NLP-oriented summarization to support faster monitoring of Big Data ethics and regulations.
- Planned reporting and alerting workflows through a web interface/API approach to generate structured outputs such as reports, newsletters, and real-time notifications.

Northwind ETL and BI Dashboard with Power BI and Talend

2025 – 2026

Academic Business Intelligence Project

USTHB

- Built a decision-support data warehouse and interactive BI dashboard based on Northwind data consolidated from SQL Server and Microsoft Access sources.
- Reimplemented in Power BI Desktop the same ETL workflow previously developed in Talend, covering extraction, cleaning, source traceability, APPEND consolidation, deduplication, and dimensional enrichment.
- Used Power Query, data modeling, relationships, and DAX measures to analyze sales performance, delivery status, customers, employees, time evolution, and territories.
- Produced analytical reports and dashboards with Power BI visualizations and Python-supported analysis to support multidimensional business insights.

University Analytics Dashboard - PRAVAN

Academic Web Dashboard Project

2024 – 2025
USTHB

- Developed a dynamic university dashboard using Python/Flask, JavaScript, AJAX REST services, and MySQL to visualize student results data.
- Built data-driven charts for decision-makers, including student counts by year, specialty evolution, gender distribution, average grades, and performance indicators.
- Connected the dashboard to the `db_university` MySQL database and analyzed the `resultats` table covering 1,336 student records from 2019 to 2021.
- Designed an interactive web interface with refresh actions, menus, and multiple visualization axes to present university data clearly.

TSP Heuristic with Kruskal and Hamiltonian Cycle Generation

Academic C++ Algorithm Project

2024 – 2025
USTHB

- Implemented a C++ heuristic for the Traveling Salesman Problem using random metric graph generation, Kruskal’s minimum spanning tree, DFS traversal, and Hamiltonian cycle construction.
- Generated distance matrices and exported results including Hamiltonian cycles, total cycle length, initial graph DOT files, and Kruskal tree DOT files for visualization.
- Tested the approach on graph sizes of 10, 20, and 50 nodes while measuring execution time and automating graph image generation with Graphviz scripts.

E-commerce Parcel Management Web Application

Bachelor’s Graduation Project

Feb. 2024 – Jun. 2024
USTHB / Zeon Soft Algeria

- Completed the application as the final Bachelor’s graduation project in Information Systems and Software Engineering.
- Developed an e-commerce parcel management web application with location-based tracking features.
- Worked on front-end and back-end components, database integration, and user workflows.
- Focused on improving delivery follow-up and parcel visibility through geolocation features.

Technical Skills

Programming Languages	Python, C++, Java, JavaScript, PHP, HTML, CSS, CPLEX, Linux programming
Data, SQL and BI	SQL, Teradata SQL, MySQL, PostgreSQL, SQL Plus, Power BI, Power Query, DAX, Talend, Tableau, Chart.js, dashboard design, query optimization
AI and Big Data	RAG pipelines, LLM-based SQL generation, prompt engineering, NLP, automated summarization, predictive analytics, ETL, data warehousing, data pipelines
Frameworks and Libraries	Flask, Django, Laravel, Streamlit, FastAPI, React Native, AJAX, BeautifulSoup, Scrapy, Selenium
APIs and Security	REST APIs, Postman, OAuth, JWT, encryption
Software Engineering	Software architecture, UML modeling, SOLID principles, design patterns, Agile/Scrum, graph algorithms, Kruskal heuristic
Tools	Visual Studio Code, Eclipse, CodeBlocks, Git, GitHub, Trello, Notion, IceScrum, Figma, Graphviz

Languages

Arabic	Native
French	Native
English	Fluent in reading, writing, and speaking

Strengths

Technical problem solving	Team collaboration	Effective communication
Responsibility	Punctuality	Adaptability